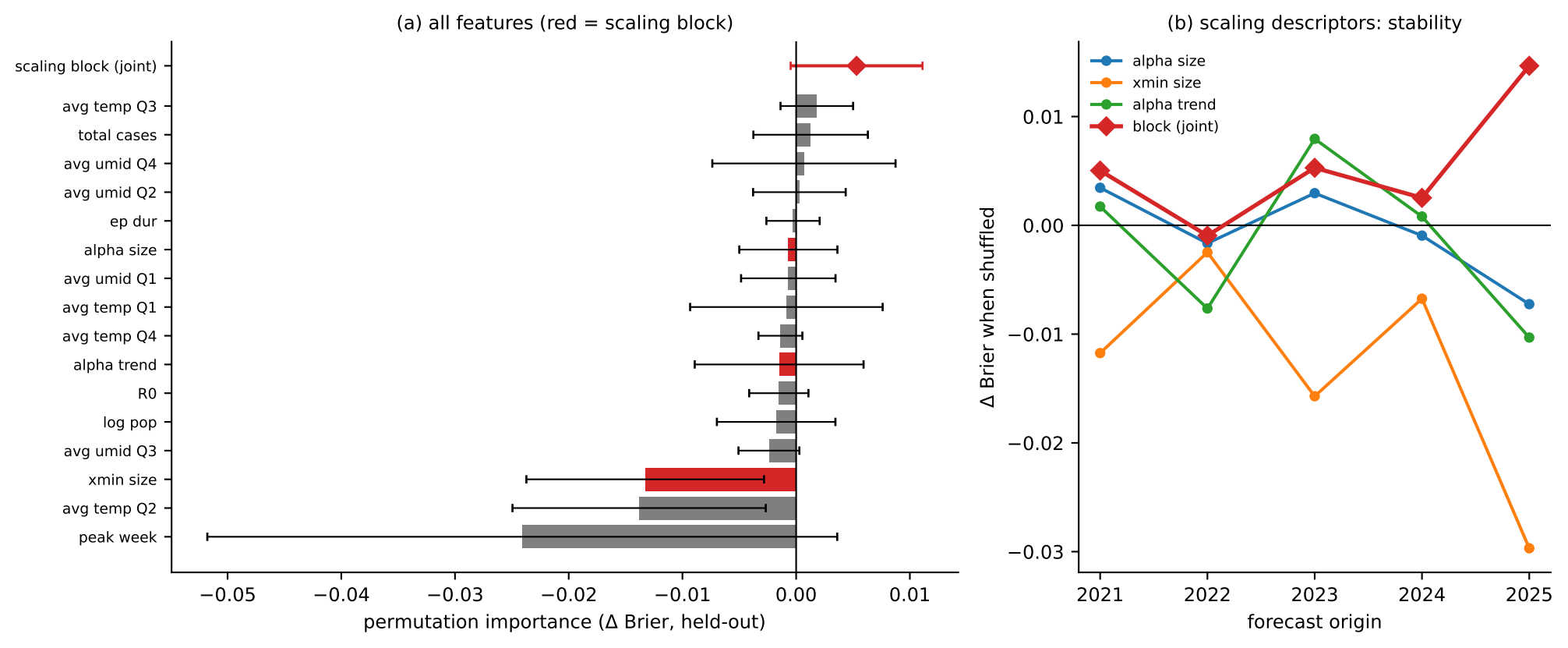
(a) all features (red = scaling block)
scaling block (joint)
avg temp Q3
total cases
avg umid Q4
avg umid Q2
ep dur
alpha size
avg umid Q1
avg temp Q1
avg temp Q4
alpha trend
R0
log pop
avg umid Q3
xmin size
avg temp Q2
peak week
−0.05
−0.04
−0.03
−0.02
−0.01
0.00
0.01
permutation importance (Δ Brier, held-out)
(b) scaling descriptors: stability
alpha size
xmin size
alpha trend
block (joint)
0.01
0.00
−0.01
−0.02
−0.03
Δ Brier when shuffled
2021
2022
2023
2024
2025
forecast origin